FA 5 – Probability and Distributions

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https://github.com/amyahespanol/Amyah-Espanol-Statistical-Theory-APM1111.git

1 Problem 1

Draw three cards in succession from a deck of 52 cards. Let:

- E_1 = King on the first draw
- E_2 = King on the second draw
- E_3 = King on the third draw

1.1 (a) $P(\overline{E_1 E_2})$

The expression $\overline{E_1 E_2}$ gives the probabilty of the intersection where at least ONE of the first two draws are NOT a king.

1.2 (b) $P(E_1 + E_2)$

The symbol $E_1 + E_2$ is the union $E_1 \cup E_2$. $P(E_1 + E_2)$ is the probability that a king is drawn on the first draw, OR on the second draw, OR on both.

1.3 (c) $\overline{E_1 + E_2}$

This is the event that neither the first nor the second draw is a king.

1.4 (d) $P(E_3 \mid E_1 \ \overline{E_2})$

This represents the conditional probability that a king is drawn on the third draw, given that the first draw was a king and the second one was NOT.

1.5 (e) $\overline{E_1 E_2 E_3}$

This gives the intersection $\overline{E_1} \cap \overline{E_2} \cap \overline{E_3}$, so a king is NOT drawn on all three draws.

1.6 (f) $P(E_1 E_2 + E_2\ \overline{E_2})$

This represents the probability that, A king is drawn on both the first and second draws OR A king is NOT drawn on the second draw and a king is drawn on the third.

2 Problem 2

2.1 (a) Probability Distribution of X

Three fair dice are rolled. Let X be the sum of the three upper faces.

The dice are independent, giving it 216 total outcomes.

$$\text{Total number of outcomes} = 6^3 = 216.$$

To determine the probability distribution of X, I used R.

```r
# 1. Generate all 216 possible outcomes for three dice
dice_grid <- expand.grid(d1 = 1:6, d2 = 1:6, d3 = 1:6)

# 2. Calculate the sum (X) for each outcome
dice_grid$X <- rowSums(dice_grid)

# 3. Create a frequency table
freq_data <- as.data.frame(table(dice_grid$X))
colnames(freq_data) <- c("x", "Frequency")

# 4. Calculate probabilities
freq_data$Probability <- as.numeric(freq_data$Frequency) / 216

# 5. Markdown table
knitr::kable(freq_data,
             col.names = c("Sum (x)", "Frequency", "P(X = x)"),
             caption = "Probability Distribution of the Sum of Three Dice",
             digits = 4)
```

Table 1: Probability Distribution of the Sum of Three Dice

Sum (x)	Frequency	P(X = x)
3	1	0.0046
4	3	0.0139
5	6	0.0278
6	10	0.0463
7	15	0.0694
8	21	0.0972
9	25	0.1157
10	27	0.1250
11	27	0.1250
12	25	0.1157
13	21	0.0972
14	15	0.0694
15	10	0.0463
16	6	0.0278
17	3	0.0139
18	1	0.0046

Therefore, the probability distribution of X is:

$$P(X = x) = \frac{\text{Number of outcomes that sum to } x}{216},$$

where $x = 3, 4, \ldots, 18$.

The distribution has symmetry around 10.5.

2.2 (b) Compute $P(7 \leq X \leq 11)$

Compute:

$$P(7 \leq X \leq 11) = P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10) + P(X = 11).$$

```
# Compute probability

prob_7_11 <- sum(freq_data$Probability[as.numeric(as.character(freq_data$x)) >= 7 & as.

prob_7_11
```

```
## [1] 0.5324074
```

Therefore,

$$P(7 \leq X \leq 11) = 0.5324074.$$

3 Problem 3

Given distribution find these probabilities.

X <- c(-10, -20, 30) p <- c(1/5, 3/10, 1/2)

3.1 (a) E(X)

EX <- sum(X * p) = 7

3.2 (b) E(X^2)

EX2 <- sum((X^2) * p) = 590

3.3 (c) Variance = E(X^2) - (E(X))^2

VarX <- EX2 - EX^2 = 541

3.4 (d) E(X^3)

EX3 <- sum((X^3) * p) = 10900

4 Problem 4

In how many ways can 3 men and 3 women be seated at a round table if:

Let the six people be distinct individuals.

```
# (a) No restriction:
# Number of circular permutations of 6 different people = (6-1)!
a <- factorial(6-1)

# (b) Two particular women must not sit together:
# Total arrangements - arrangements where W1 and W2 sit together
total <- a
# keep W1 and W2 as one -> 5 people around a circle -> (5-1)!; multiply by 2
together <- factorial(5-1) * 2
not_together <- total - together

# (c) Each woman is to be between two men (men and women alternate).
# Number of alternating seatings, men first around the circle: (3-1)!
```

```r
# Place the 3 distinct women in the 3 gaps: 3!
alternating <- factorial(3-1) * factorial(3)

list(
  a_no_restriction = a,
  b_not_together = not_together,
  c_alternating = alternating
)
```

```
## $a_no_restriction
## [1] 120
##
## $b_not_together
## [1] 72
##
## $c_alternating
## [1] 12
```

4.1 (a)

Number of arrangements with no restriction: 120

4.2 (b)

Number of arrangements where two particular women are not together: 72

4.3 (c)

Number of alternating seatings (each woman between two men): 12

5 Problem 5

In how many ways can 2 men, 4 women, 3 boys, and 3 girls be selected from:
6 men, 8 women, 4 boys, 5 girls

```r
choose <- choose

# (a) No restrictions
a_men <- choose(6,2)
a_women <- choose(8,4)
```

```r
a_boys <- choose(4,3)
a_girls <- choose(5,3)
total_a <- a_men * a_women * a_boys * a_girls

# (b) A particular man and a particular woman must be selected.

# For men: 2 men needed total, 1 is fixed -> choose 1 from remaining 5
b_men <- choose(5,1)
# For women: 4 women needed total, 1 is fixed -> choose 3 from remaining 7
b_women <- choose(7,3)
b_boys <- a_boys
b_girls <- a_girls
total_b <- b_men * b_women * b_boys * b_girls

list(
  total_a = total_a,
  total_b = total_b
)
```

```
## $total_a
## [1] 42000
##
## $total_b
## [1] 7000
```

5.1 (a)

No restriction: 4.2×10^4

5.2 (b)

A particular man and a particular woman must be selected: 7000

6 Problem 6

Prove that for any two events E_1 and E_2,

$$P(E_1 + E_2) \leq P(E_1) + P(E_2)$$

6.1 Solution

Using the Addition Rules for probabilities:

$$P(E_1 + E_2) = P(E_1) + P(E_2) - P(E_1E_2)$$

Since the probability of an intersection can't be negative,

$$P(E_1E_2) \geq 0$$

Subtracting a nonnegative number can't make the result larger, so

$$P(E_1 + E_2) = P(E_1) + P(E_2) - P(E_1E_2) \leq P(E_1) + P(E_2)$$

Therefore,

$$\boxed{P(E_1 + E_2) \leq P(E_1) + P(E_2)}$$

Conclusion:
The probability of the union of two events is always less than or equal to the sum of their probabilities.